SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

CHAPTER 10

Human Eye and Colourful World

(Full Chapter)

> Structure of Human Eye • Power of Accommodation • Defects of Vision & Correction • Refraction Through Glass Prism • Dispersion of Light • Atmospheric Refraction • Scattering of Light

Target Level: CBSE Class 10

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📖 Chapter Overview & Basic Concepts

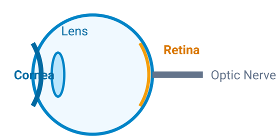
The human eye is a natural optical instrument that uses a convex lens to form a real and inverted image of objects on a light-sensitive screen called the retina.

SECTION 1: THE HUMAN EYE & ACCOMMODATION

1.1 Main Parts of the Human Eye

- **Cornea:** Transparent, spherical membrane at the front of the eye; majority of light refraction occurs at its outer surface.
- **Iris:** Dark muscular diaphragm behind the cornea that controls the size of the pupil.
- **Pupil:** Central aperture through which light enters the eye; appears black because no light reflects back.
- **Ciliary Muscles:** Hold the crystalline lens in position and modify its curvature to change focal length.
- **Eye Lens:** Fibrous, jelly-like convex lens that forms a real and inverted image on the retina.
- **Retina:** Light-sensitive inner lining containing photoreceptor cells (rods for light intensity and cones for color vision).
- **Optic Nerve:** Transmits electrical signals from photoreceptors to the brain.

📌 **Diagram: Structure of Human Eye**



1.2 Power of Accommodation

The ability of the eye lens to adjust its focal length using ciliary muscles to view both near and distant objects clearly.

- **Viewing Distant Objects:** Ciliary muscles relax → Lens becomes thin → Focal length increases.
- **Viewing Nearby Objects:** Ciliary muscles contract → Lens becomes thick & rounded → Focal length decreases.
- **Near Point (Least Distance of Distinct Vision):** Minimum distance for clear vision without strain = **25 cm** for a normal young adult.
- **Far Point:** Maximum distance up to which the eye can see objects clearly = **Infinity (∞)**.

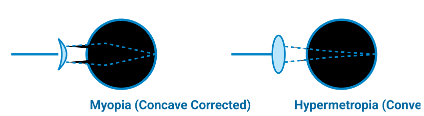
> **NCERT High-Yield Insight:** Cataract is a condition in old age where the crystalline lens becomes milky and cloudy due to membrane growth. It causes partial or complete loss of vision and can be restored only through cataract surgery.

SECTION 2: DEFECTS OF VISION AND THEIR CORRECTION

Defect	Description	Causes	Corrective Lens
Myopia (Near-Sightedness)	Can see nearby objects clearly, but cannot see distant objects distinctly. Far point is closer than infinity. Image forms *in front of retina*.	1. Excessive curvature of eye lens. 2. Elongation of the eyeball.	**Concave Lens** of suitable power.
Hypermetropia (Far-Sightedness)	Can see distant objects clearly, but cannot see nearby objects distinctly. Near point is farther than 25 cm. Image forms *behind retina*.	1. Focal length of eye lens is too long. 2. Eyeball has become too small.	**Convex Lens** of suitable power.
Presbyopia (Old-Age Hypermetropia)	Power of accommodation decreases with aging. Hard to see nearby objects comfortably.	1. Gradual weakening of ciliary muscles. 2. Diminishing flexibility of eye lens.	**Bifocal Lens** (Upper part concave, lower part convex).

Important Lens Power Formulae:
- Power of Corrective Lens: $P = \frac{1}{f \text{ (in meters)}}$
- Lens Formula: $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

📌 **Diagram: Correction for Myopic and Hypermetropic Eye**



SECTION 3: REFRACTION & DISPERSION THROUGH A PRISM

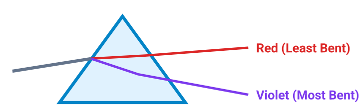
3.1 Refraction Through a Triangular Glass Prism

- When light passes through a glass prism, it bends towards the base at the first surface and bends further towards the base at the second surface.
- **Angle of Deviation ($\angle D$):** The angle between the direction of incident ray and emergent ray.

3.2 Dispersion of White Light

- **Dispersion:** The splitting of white light into its component seven colors when passing through a transparent prism.
- **Spectrum Colors: VIBGYOR** (Violet, Indigo, Blue, Green, Yellow, Orange, Red).
- **Cause:** Different wavelengths bend through different angles relative to the incident ray. Red light (longest λ) bends the least; Violet light (shortest λ) bends the most.

📌 **Diagram: Dispersion of White Light Through Glass Prism**



3.3 Recombination & Rainbow Formation

- **Recombination:** Isaac Newton showed that placing an inverted second prism next to the first recombines the seven spectrum colors back into white light.
- **Rainbow Formation:** A natural spectrum caused by dispersion of sunlight by tiny water droplets in the atmosphere.

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Sunlight enters raindrop → Refraction & Dispersion → Total Internal Reflection →
Refraction out of droplet → Rainbow seen
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SECTION 4: ATMOSPHERIC REFRACTION & SCATTERING OF LIGHT

4.1 Atmospheric Refraction Phenomena

Refraction of light caused by earth's atmosphere due to varying optical densities of air layers at different temperatures.

- **Twinkling of Stars:** Starlight passes through atmospheric layers with continuously fluctuating refractive indices. The apparent position fluctuates, making the star appear to twinkle.
- **Why Planets Don't Twinkle:** Planets are closer to Earth and act as extended sources (a collection of point sources). Fluctuations average out to zero.
- **Advanced Sunrise & Delayed Sunset:** The Sun is visible ~2 minutes before actual sunrise and ~2 minutes after actual sunset due to atmospheric bending of light.

4.2 Scattering of Light & Applications

- **Tyndall Effect:** Path of light becomes visible when passing through a colloidal solution due to scattering by fine particles (e.g., sunlight passing through dense forest canopy).
- **Blue Color of Sky:** Fine particles in air scatter light of shorter wavelengths (blue end) much more strongly than longer wavelengths (red end).
- **Red Appearance of Sun at Sunrise/Sunset:** Light travels a longer distance through atmosphere. Shorter blue light scatters away; mostly unscattered longer red light reaches our eyes.
- **Danger Signal Lights are Red:** Red light is scattered least by fog and smoke, so it remains visible from the maximum distance.

❓ Important Practice Questions

Q1: What is the power of accommodation of the human eye? State its value for a normal eye.
Ans: It is the ability of the eye lens to adjust its focal length using ciliary muscles to focus on near and far objects. For a normal adult eye, the range of vision is 25 cm to infinity.

Q2: A person needs a lens of power -2.5 D to correct his vision. Identify the defect and calculate the focal length of the corrective lens.
Ans: Negative power indicates a **Concave Lens**, so the defect is **Myopia**.
Focal Length $f = \frac{1}{P} = \frac{1}{-2.5 \text{ D}} = -0.4 \text{ m} = \mathbf{-40 \text{ cm}}$.

Q3: Why does clear sky appear blue while to an astronaut in space it appears dark?
Ans: Atmosphere scatters shorter blue light toward Earth. In space, there is no atmosphere to scatter light, so no scattered light enters the astronaut's eye, making space appear dark.

Q4: Explain why planets do not twinkle like stars.
Ans: Planets are closer and appear as extended sources of light. The total variation in light intensity from all individual point sources averages out to zero, nullifying the twinkling effect.